CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Gases such as neon and argon are used in lamps because they are: 1

(a) magnetic **(b)** reactive **(c)** heavy **(d)** chemically inert

2. A substance which on electrolysis gives hydrogen and oxygen is: 1

(a) air **(b)** salt **(c)** water **(d)** sugar

3. Which of the following is an example of a molecule of an element? 1

(a) P_4 **(b)** CO **(c)** H_2O **(d)** NH_3

4. Sodium is stored under kerosene because it: 1

(a) dissolves in kerosene **(b)** is a gas

(c) reacts vigorously with air and water **(d)** is light

5. Which gas is formed on burning carbon in air? 1

(a) carbon dioxide **(b)** hydrogen **(c)** oxygen **(d)** nitrogen

6. Which of the following can be broken into simpler substances only by chemical means? 1

(a) iron **(b)** gold **(c)** water **(d)** oxygen

7. The water of the sea is separated into salt and water by: 1

(a) winnowing **(b)** evaporation and condensation

(c) sieving **(d)** magnet

8. A mixture of oil and water can be separated by: 1

(a) sublimation **(b)** magnet

(c) sieving **(d)** separating funnel

9. A student mixes sugar and sand. To separate them, the best method is: 1

(a) only sieving **(b)** dissolve in water, filter, evaporate

(c) sublimation **(d)** magnet

10. Diamond and graphite are both forms of: 1

(a) silicon **(b)** sulphur **(c)** carbon **(d)** oxygen

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why it is not possible to separate the components of a compound by physical methods. 2

12. Why are alloys more useful than pure metals? Give an example. 2

13. Differentiate between a molecule of an element and of a compound with examples. 2

14. Name two elements whose compounds are present in common salt and baking soda. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Write the number of atoms and the elements in H_2SO_4, NH_3 and CH_4. 3

16. Describe how you will separate a mixture of salt and sand and one of sulphur and iron. 3

17. How do you show that air is a mixture and not a compound? 3

OR

Explain the use of non-metals in each: 3
(a) chlorine
(b) sulphur
(c) nitrogen.

18. State three differences between a metal and a metalloid with examples. 3

OR

Explain how bronze statues and bells are manufactured by using properties of alloys. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain why alloys are preferred. Discuss steel, stainless steel, brass and bronze with their composition and uses. 5

OR

Explain the properties of the three allotropes of carbon: diamond, graphite and charcoal. Why are they considered one element? 5

20. Explain atoms and molecules with examples of H, H2, O2, H2O, CO2 and NH3. State the number of atoms in each molecule. 5

OR

Identify the type of mixtures and suggest the separation method: 5
(a) chalk powder in water
(b) ink in water
(c) kerosene and water
(d) salt and ammonium chloride
(e) mustard seeds and wheat.

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND